5.7.4 Unification

Unification is the process of solving equations between symbolic expressions. One of the ways of solving such an equation is finding terms on both sides that are equivalent, and removing them from the equation.

Of course, solution strategies depend on which expressions (terms) can appear in the equation and how you can compare them or when are they considered to be equal. For instance, they might be evaluated and considered equal if they have the same value, or they might be symbolic and considered equal if they are equivalent, possibly net of some variable substitution.

As you can imagine, the disjoint set data structure is perfect for high-performance algorithms solving this problem.

5.8 Summary

In this chapter, we have seen how even apparently simple problems can be underestimated, turning into performance paint points. This was the perfect example of how we can use proper analysis and algorithms basics to tackle problems.

- **Cleanness**: the beauty of disjoint-set is that we can build increasingly complex and efficient solutions to solve it, just adding small incremental changes. This makes comparing the solutions and understanding the differences between them easier.